King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 208 Major Exam I Term 241

Original exam date: 1 October 2024. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. A general solution of the differential equation

$$\frac{dy}{dx} = (x + y + 1)^2$$

is

(a) $y = -x + 1 + \sec(x + c)$

(b) $y = -x + 1 + \tan(x + c)$

(c) $y = -x - 1 + \tan(x + c)$

(d) $y = x + 1 + \tan(x + c)$

(e) $y = -x - 1 + \sec(x + c)$

2. A general solution of the differential equation

$$x^2y'' + 2xy' = 12x^3$$

is

(a) $y(x) = x^3 + \dfrac{A}{x} + B$

(b) $y(x) = x^2 - \dfrac{A}{x} + B$

(c) $y(x) = x^2 + \dfrac{A}{x} + B$

(d) $y(x) = x^3 - \dfrac{A}{x} + B$

(e) $y(x) = x^3 - Ax^2 + B$

3. The general solution of the linear differential equation

$$2xy' - 3y = 9x^3$$

is given by

(a) $y(x) = x^3 + cx^{\frac{2}{3}}$

(b) $y(x) = x^2 + cx^{\frac{3}{2}}$

(c) $y(x) = 3x^3 + cx^{\frac{3}{2}}$

(d) $y(x) = 3x^2 + cx^{\frac{-3}{2}}$

(e) $y(x) = 2x^3 + cx^{\frac{5}{2}}$

4. Let $\mathbf{u} = (4, 5)$, $\mathbf{v} = (-2, 7)$, $\mathbf{w} = (8, 29)$ be vectors in $\mathbb{R}^2$. If $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$, then $a - b =$

(a) 0

(b) 2

(c) 1

(d) 3

(e) 4

5. A general solution of the exact differential equation

$$\left(3x^2 + 2y^2\right) dx + \left(4xy + 6y^2\right) dy = 0$$

is

(a) $x^3 - 2xy^2 + 2y^3 = C$

(b) $x^3 + xy^2 + y^3 = C$

(c) $x^3 - 2xy^2 - 2y^3 = C$

(d) $x^3 + 2xy^2 - 2y^3 = C$

(e) $x^3 + 2xy^2 + 2y^3 = C$

6. By making a suitable substitution, the differential equation

$$xy' + 6y = 3xy^{\frac{4}{3}}$$

can be transformed into a linear differential equation

(a) $v' - \dfrac{2}{x}v = -1$

(b) $v' - \dfrac{2}{x}v = 1$

(c) $v' + \dfrac{2}{x}v = -1$

(d) $v' + \dfrac{2}{x}v = x$

(e) $v' - \dfrac{2}{x}v = x$

7. By making a suitable substitution, the differential equation

$$x^4y' = x^3y - 5y^4$$

can be transformed into a separable differential equation

(a) $\dfrac{dv}{v^2} + \dfrac{5}{x}\,dx = 0$

(b) $\dfrac{dv}{v^4} + \dfrac{5}{x}\,dx = 0$

(c) $\dfrac{dv}{v^3} - \dfrac{5}{x}\,dx = 0$

(d) $\dfrac{dv}{v^3} + \dfrac{5}{x}\,dx = 0$

(e) $\dfrac{dv}{v^4} - \dfrac{5}{x}\,dx = 0$

8. The solution of the system

$$\begin{aligned} x_1 - 2x_2 - 4x_3 + 8x_4 &= 0 \\ 2x_1 + 3x_2 + 6x_3 + 9x_4 &= 0 \\ 3x_1 + 5x_2 + 4x_3 + x_4 &= 0 \end{aligned}$$

is the set of all scalar multiples of a vector u where $u =$

(a) $(-6, 5, -2, 1)$

(b) $(6, 5, -2, 1)$

(c) $(-6, -5, -2, 1)$

(d) $(-6, 5, 2, 1)$

(e) $(-6, -5, -2, -1)$

9. If a certain substance cools from 100°C to 60°C in 10 minutes when it is taken outside where the air temperature is 20°C, then the temperature of the substance 40 minutes after it is taken outside is

(a) 40°C

(b) 35°C

(c) 10°C

(d) 15°C

(e) 25°C

10. The value of k for which the vectors $\mathbf{u} = (1, 4, 5)$, $\mathbf{v} = (4, 2, k)$, $\mathbf{w} = (-3, 3, -1)$ of $\mathbb{R}^3$ are linearly dependent is

(a) $\dfrac{-96}{13}$

(b) $\dfrac{96}{13}$

(c) $\dfrac{103}{15}$

(d) $\dfrac{104}{15}$

(e) 0

11. The explicit particular solution of the initial-value problem

$$x^2 \frac{dy}{dx} + y = xy, \qquad y(1) = 2$$

is $y(x) =$

(a) $2xe^x - 1$

(b) $xe^{\frac{1}{x}} + 1$

(c) $2xe^{\frac{1}{x}-1}$

(d) $2x^2e^{\frac{1}{x}-1}$

(e) $2x^2e^x - 1$

12. Which one of the following statements is TRUE about the subset V of $\mathbb{R}^3$ defined by $V = \{(x_1, x_2, x_3) : x_2 = 1\}$?

(a) V is not closed under addition and not closed under multiplication by scalar

(b) V is not closed under addition but closed under multiplication by scalar

(c) V is closed under addition and not closed under multiplication by scalar

(d) V is closed under addition

(e) V is a subspace of $\mathbb{R}^3$

13. A particle is moving in a straight line with acceleration $a(t) = 4(t+3)^2$, and an initial position $x(0) = 1$, and an initial velocity $v(0) = -1$, then the position function $x(t)$ of the particle is given by

(a) $x(t) = \frac{1}{3}(t+3)^4 - 30t - 26$

(b) $x(t) = \frac{1}{3}(t+3)^4 + 37t - 26$

(c) $x(t) = \frac{1}{4}(t+3)^4 - 37t - 26$

(d) $x(t) = \frac{1}{3}(t+3)^4 - 37t - 26$

(e) $x(t) = \frac{1}{4}(t+3)^4 + 37t - 26$

14. Find the constant A that makes the differential equation

$$\left(\frac{1}{x^2} + \frac{1}{y^2}\right) dx + \left(\frac{Ax+1}{y^3}\right) dy = 0$$

exact.

(a) 2

(b) -3

(c) 0

(d) -2

(e) 3

15. The sum of all values of r such that $y = e^{rx}$ is a solution of the differential equation

$$3y'' + 3y' - 4y = 0$$

is

(a) 1

(b) -1

(c) $-\frac{3}{2}$

(d) $\frac{1}{2}$

(e) 0

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(c)**	$v = x + y + 1$: $v' = 1 + v^2$, $v = \tan(x + c)$, so $y = -x - 1 + \tan(x + c)$.
2	**(d)**	$p = y'$: $(x^2 p)' = 12x^3$, $p = 3x^2 + \frac{A}{x^2}$, so $y = x^3 - \frac{A}{x} + B$.
3	**(c)**	$y' - \frac{3}{2x} y = \frac{9}{2} x^2$, $\rho = x^{-3/2}$: $x^{-3/2} y = 3x^{3/2} + c$.
4	**(c)**	$4a - 2b = 8$, $5a + 7b = 29$: $a = 3$, $b = 2$, so $a - b = 1$.
5	**(e)**	$F = x^3 + 2xy^2 + g(y)$, $g' = 6y^2$: $x^3 + 2xy^2 + 2y^3 = C$.
6	**(a)**	Bernoulli $n = \frac{4}{3}$, $v = y^{-1/3}$: multiply by $-\frac{1}{3} y^{-4/3}$, $v' - \frac{2}{x} v = -1$.
7	**(b)**	Homogeneous, $v = y/x$: $xv' = -5v^4$, so $\frac{dv}{v^4} + \frac{5}{x}\, dx = 0$.
8	**(a)**	Row reduce: $x_3 = -2x_4$, $x_2 = 5x_4$, $x_1 = -6x_4$, so $u = (-6, 5, -2, 1)$.
9	**(e)**	$T = 20 + 80e^{-kt}$, $e^{-10k} = \frac{1}{2}$: $T(40) = 20 + 80(\frac{1}{2})^4 = 25°\text{C}$.
10	**(d)**	$\det = 104 - 15k = 0$, so $k = \frac{104}{15}$.
11	**(c)**	Separable: $\frac{dy}{y} = (\frac{1}{x} - \frac{1}{x^2})dx$, $y = Kxe^{1/x}$, $K = \frac{2}{e}$.
12	**(a)**	x_2 of a sum is 2 and of $c\mathbf{v}$ is c: neither closure holds.
13	**(d)**	$v = \frac{4}{3}(t + 3)^3 - 37$ (from $v(0) = -1$), then $x = \frac{1}{3}(t + 3)^4 - 37t - 26$.
14	**(d)**	$M_y = -\frac{2}{y^3}$, $N_x = \frac{A}{y^3}$, so $A = -2$.
15	**(b)**	$3r^2 + 3r - 4 = 0$, roots $\frac{-3 \pm \sqrt{57}}{6}$, sum $= -1$.